

Lecture 1

Two Paradigms

Classical Statistics

- Infer information from small data sets (Not enough data)

Machine Learning

- Infer information from large data sets (Too many data)

Machine Learning

Supervised learning

Learn from labeled input–output examples.

- Classification
- Regression

Unsupervised learning

Find structure without externally supplied targets.

- Clustering
- Dimensionality reduction

Reinforcement learning

Learn actions from rewards through interaction with an environment.

Tasks

- Supervised Learning: given examples of inputs and corresponding desired outputs, predict outputs on future inputs.
e.g.: classification, regression
- Unsupervised Learning: given only inputs, automatically discover representations, features, structure, etc.
e.g.: clustering, dimensionality reduction, feature extraction

Classification

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^d, \quad y_i \in \mathcal{Y}, \quad |\mathcal{Y}| < \infty.$$

Find a function

$$f : \mathbb{R}^d \rightarrow \mathcal{Y}$$

so that for a new input x , the prediction is $f(x)$.

Regression

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathbb{R}^d, \quad y_i \in \mathbb{R}.$$

Find a function

$$f : \mathbb{R}^d \rightarrow \mathbb{R}$$

so that for a new input x , the predicted response is $f(x)$.

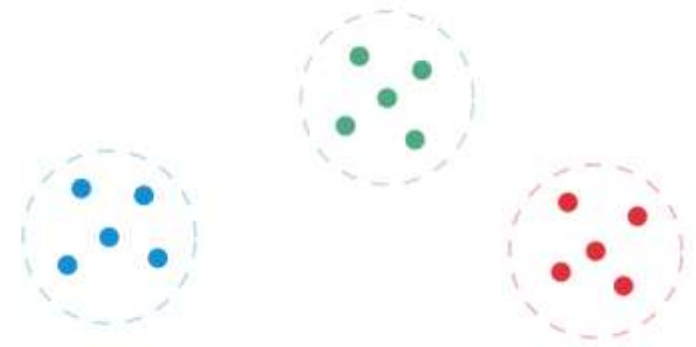
Clustering

$$\mathcal{D} = \{x_i\}_{i=1}^n, \quad x_i \in \mathbb{R}^d.$$

Find an assignment

$$f : \mathbb{R}^d \rightarrow \{1, \dots, K\}$$

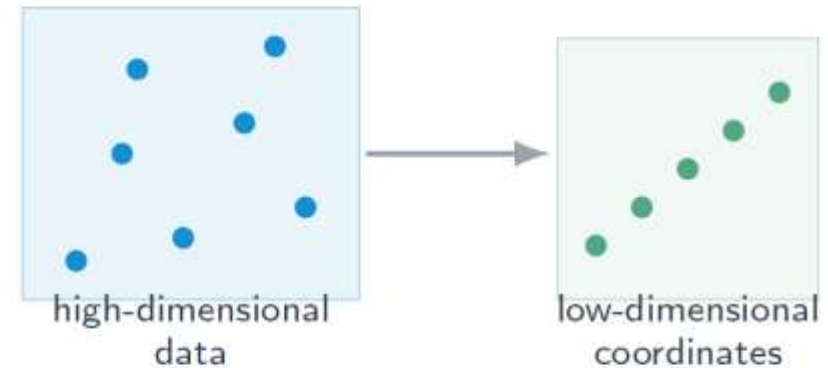
such that similar observations receive the same cluster label.



Dimensionality reduction

Represent high-dimensional data with fewer coordinates while preserving useful structure.

- Examples: PCA, manifold learning, autoencoders
- Uses: visualization, compression, denoising, features



Beyond the classical four

Adapting with limited examples

- Zero-shot learning
- One-shot and few-shot learning
- In-context learning

Learning across tasks and time

- Transfer learning
- Multi-task learning
- Meta-learning
- Continual learning

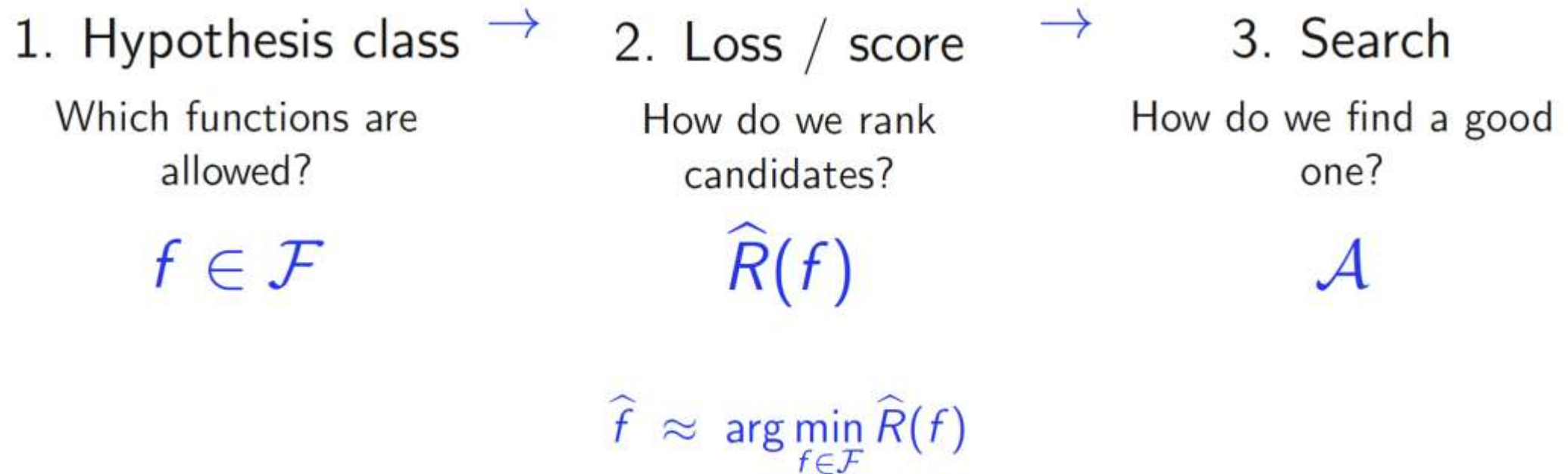
These regimes do not replace the classical task types.

They describe how a model transfers, adapts, or continues learning across tasks.

What is Machine Learning?

- Machine Learning is just glorified "curve fitting"?

Three Ingredients of a Learning Algorithm



A method changes when its candidate set, ranking rule, or search strategy changes.

1. Hypothesis Class: The Candidate Set

The hypothesis class is the set of candidate prediction rules.

$$\mathcal{F} = \{f_{\theta} : \theta \in \Theta\}$$

- ♦ **Linear rules:** Hyperplanes, logistic regression, perceptrons, and linear SVMs.
- ♦ **Structured nonlinear rules:** Trees, kernels, nearest neighbours, and graphical models.
- ♦ **Neural networks:** Feedforward networks, CNNs, and other learned representations.

If $\mathcal{F}$ cannot represent a suitable rule, a better optimizer cannot remove that limitation.

2. Loss: Ranking Candidate Models

A per-example loss measures the cost of a prediction:

$$\ell(y, f(x)).$$

The empirical risk aggregates those costs:

$$\hat{R}(f) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)).$$

The loss therefore induces a ranking:

$$f_a \text{ is preferred to } f_b \iff \hat{R}(f_a) < \hat{R}(f_b).$$

The same hypothesis class can produce a different classifier when the loss changes.

- ♦ **0–1 loss:** ranks by classification errors.
- ♦ **Log loss:** rewards accurate probabilities.
- ♦ **Hinge loss:** rewards large margins.

3. Search: Finding a Preferred Model

Once candidates can be ranked, an algorithm must search for a good one.

$$\hat{f} = \mathcal{A}(\mathcal{F}, \hat{R}, \mathcal{D}) \approx \arg \min_{f \in \mathcal{F}} \hat{R}(f).$$

- ◆ **Closed form:** Linear algebra or a special analytic solution.
- ◆ **Convex optimization:** Numerical search with a global optimum.
- ◆ **Gradient methods:** SGD and its variants for neural networks.
- ◆ **Greedy or combinatorial search:** Trees, feature selection, and structures.

Search may be exact or approximate, explicit or implicit; it selects the final model.

Applications of Classification in Healthcare

- 1.Disease Prediction:** Classifying patients based on the likelihood of developing a particular disease.
- 2.Drug Classification:** Categorizing drugs based on their effectiveness for different patient groups.
- 3.Medical Imaging:** Identifying abnormalities in X-rays, MRIs, and other medical images.

Applications of Classification in Finance

- 1.Credit Scoring:** Classifying the creditworthiness of individuals.
- 2.Fraud Detection:** Identifying fraudulent transactions in real-time.
- 3.Customer Segmentation:** Classifying customers into different segments based on spending habits.

Applications of Classification in Technology

- 1.Spam Filtering:** Classifying emails as spam or not spam.
- 2.Sentiment Analysis:** Categorizing opinions expressed in text as positive, negative, or neutral.
- 3.Object Recognition:** Identifying objects within images or video feeds.

Applications of Classification in Marketing

- 1.Customer Churn Prediction:** Identifying customers who are likely to leave a service.
- 2.Market Basket Analysis:** Classifying items that are likely to be purchased together.
- 3.Ad Targeting:** Classifying users to show them relevant advertisements.

Application of Classification in Autonomous Vehicles

1. **Pedestrian Detection:** Real-time classification of objects as pedestrians, vehicles, or other obstacles.
2. **Traffic Sign Recognition:** Classifying traffic signs to inform the vehicle's decision-making.
3. **Lane Departure Warning:** Classifying road lanes and alerting the driver or taking corrective action if the vehicle departs from its lane.
4. **Adaptive Cruise Control:** Classifying vehicles on the road to maintain a safe following distance.

Data



A Three-Class Classification



African savanna elephant

African forest elephant

Asian elephant

Feature vector (x)

Illustrative features (0–100): ear ratio, tusk angle, head-profile score

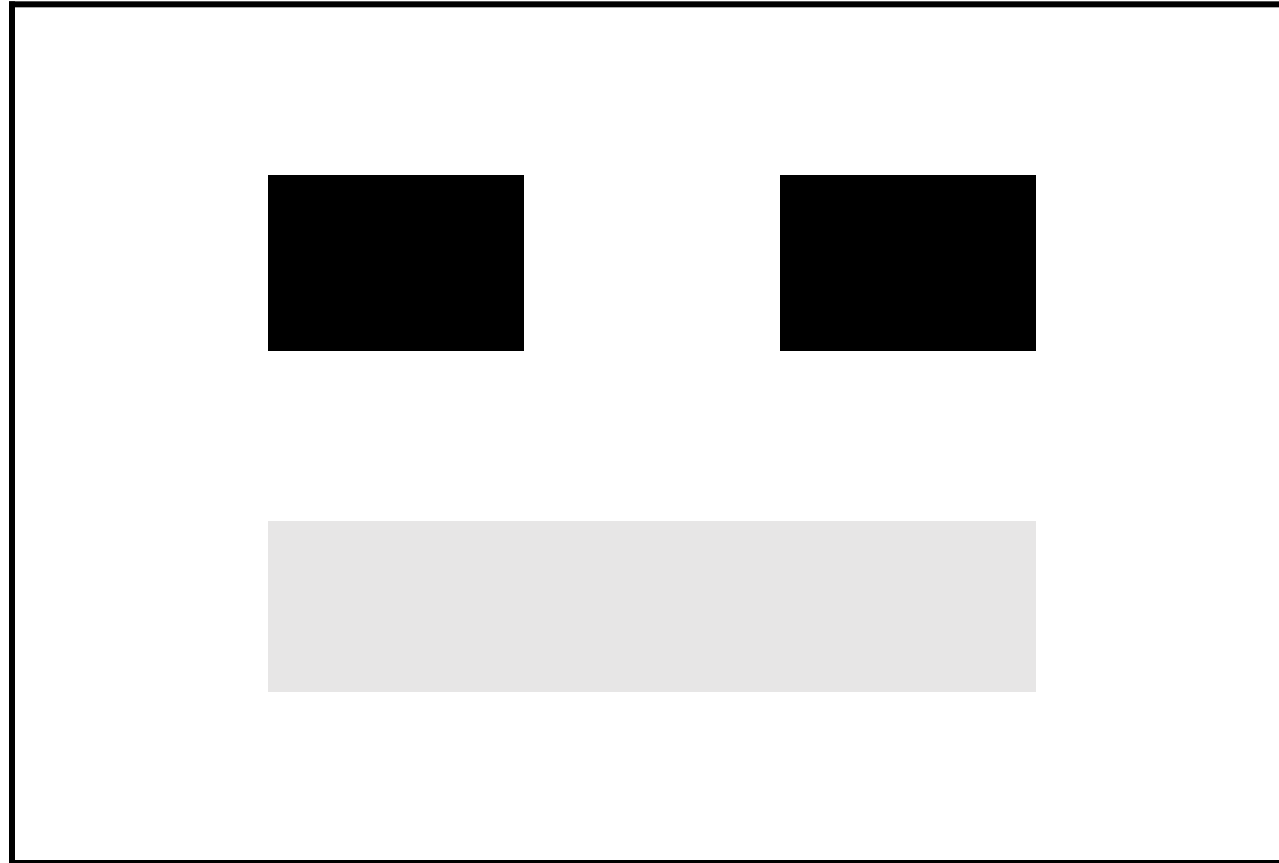


African Savanna Elephant
 $x_1 = (85, 42, 15)$

African Forest Elephant
 $x_2 = (68, 15, 20)$

Asian Elephant
 $x_3 = (35, 55, 90)$

Data Representation



Data Representation

Data Representation

1	1	1	1	1
1	0	1	0	1
1	1	1	1	1
1	0.5	0.5	0.5	1
1	1	1	1	1

From Objects to Labels

Elephant images $\longrightarrow$ features X $\longrightarrow$ species Y



Classifying a New Elephant



$$x = (82, 40, 18)$$

$$f(82, 40, 18) = ?$$

Classifying a New Elephant



$$x = (65, 18, 22)$$

$$f(65, 18, 22) = ?$$

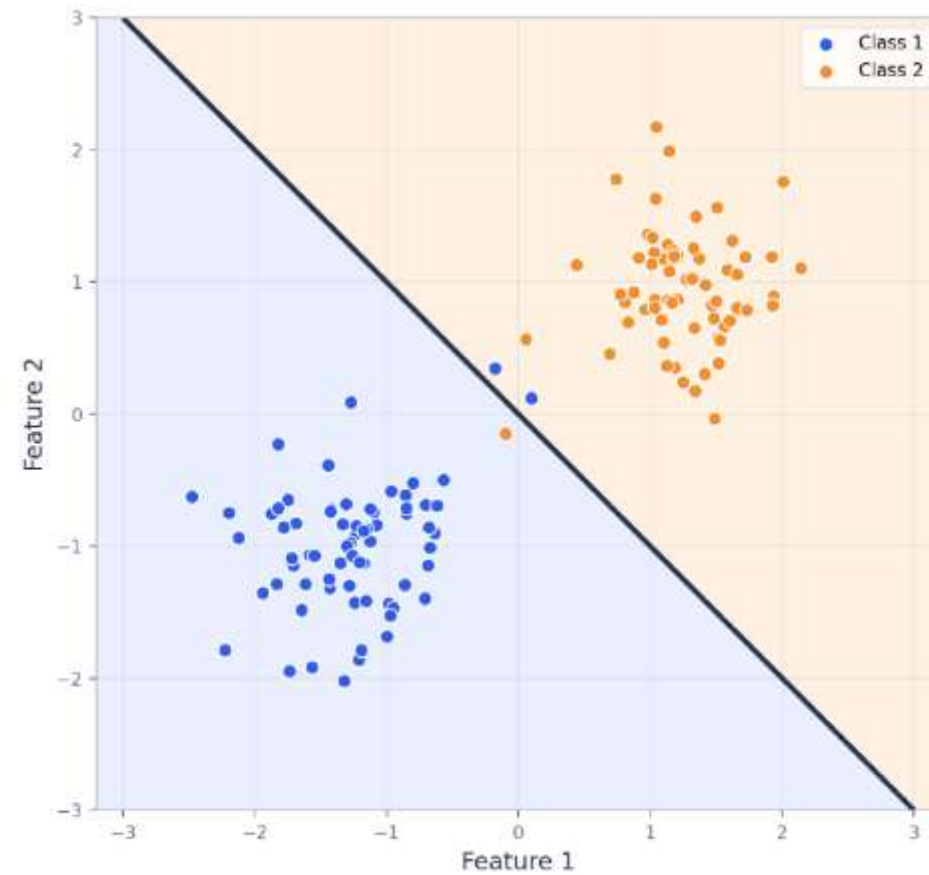
Classifying a New Elephant



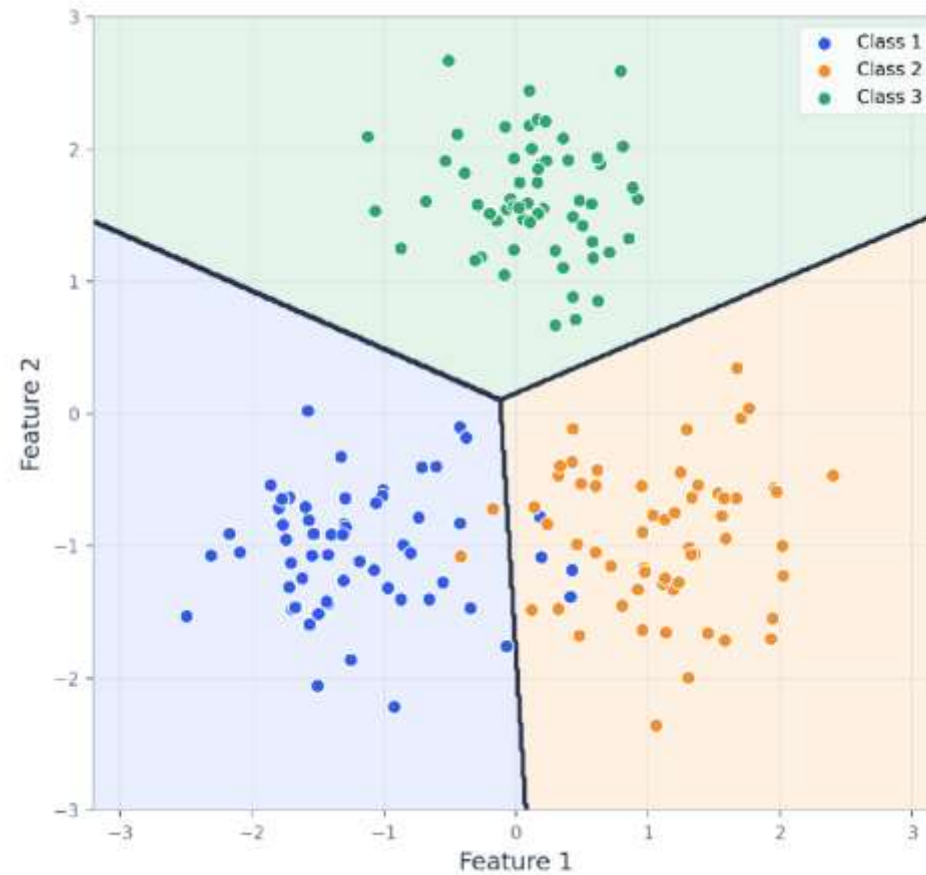
$$x = (38, 52, 88)$$

$$f(38, 52, 88) = ?$$

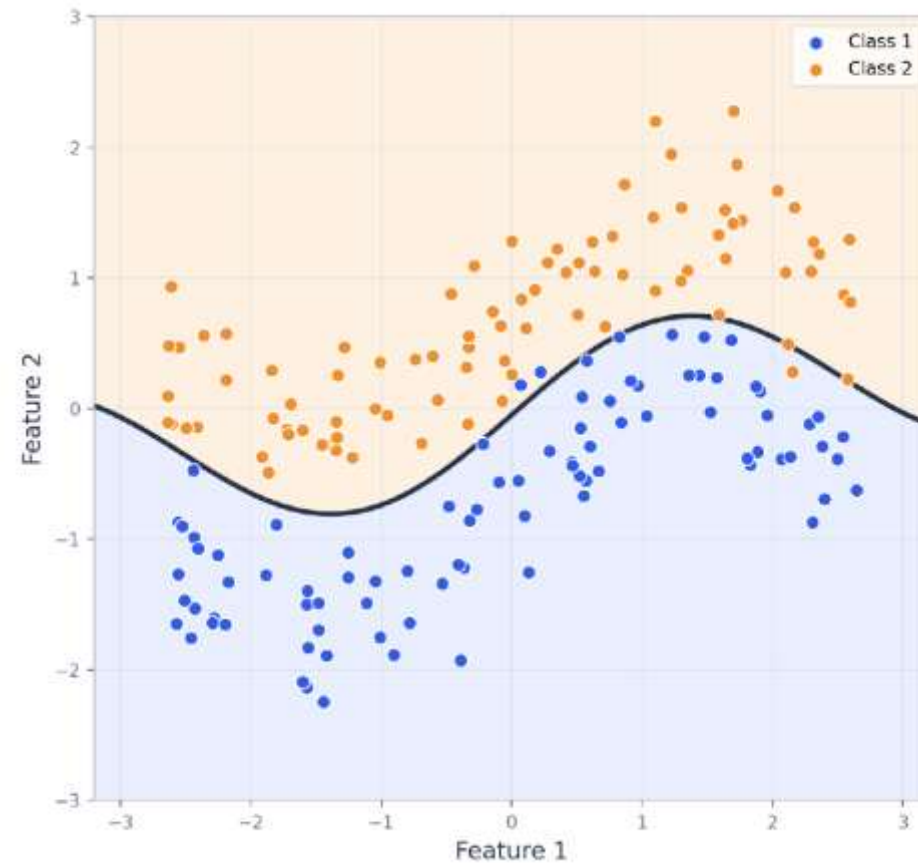
Linear Classification



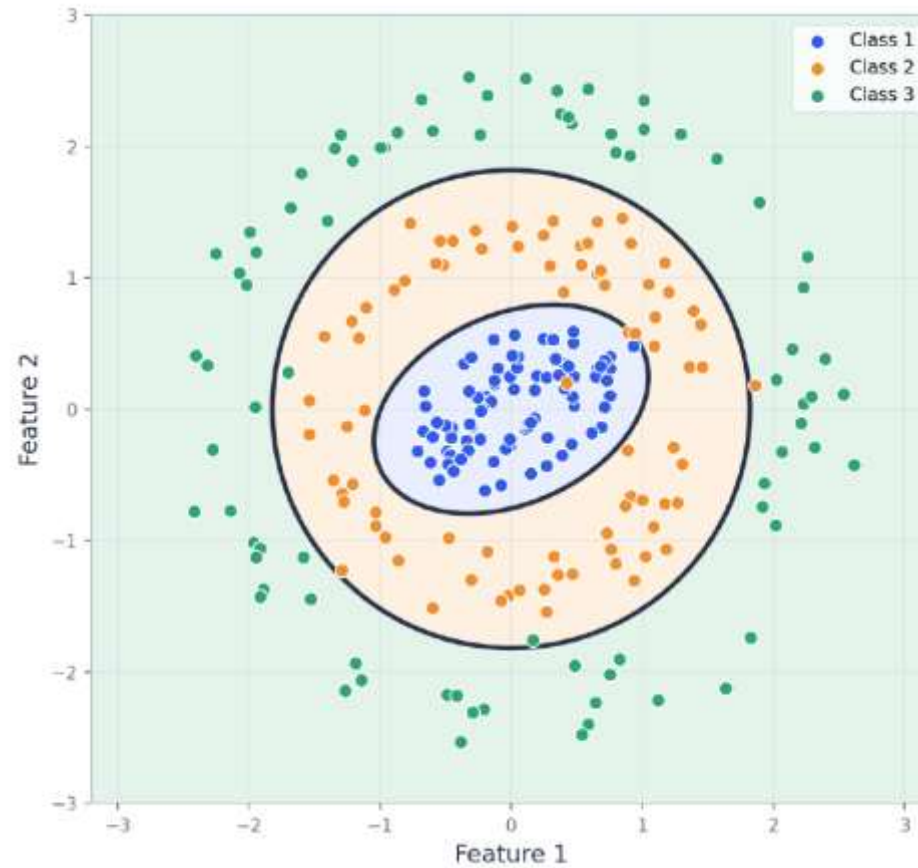
Multiclass Classification



Nonlinear Classification



Nonlinear Multiclass Classification STAT



Digit Recognition

The Matlab data file 2_3.mat contains 200 handwritten 2's and 3's images scanned from postal envelopes, like the ones shown below.

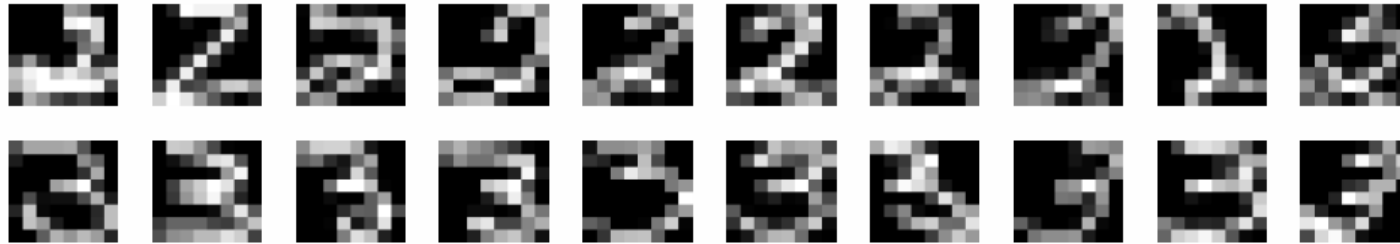
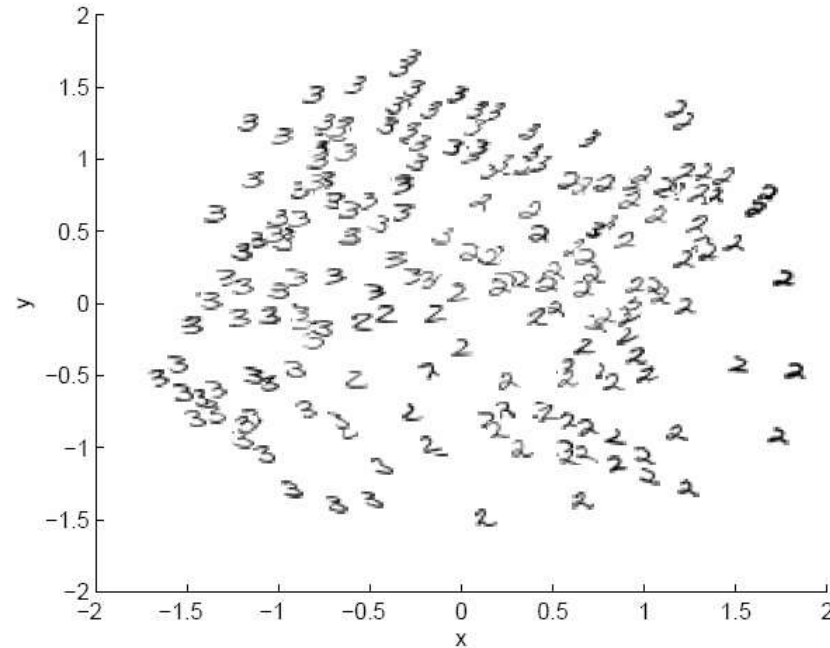


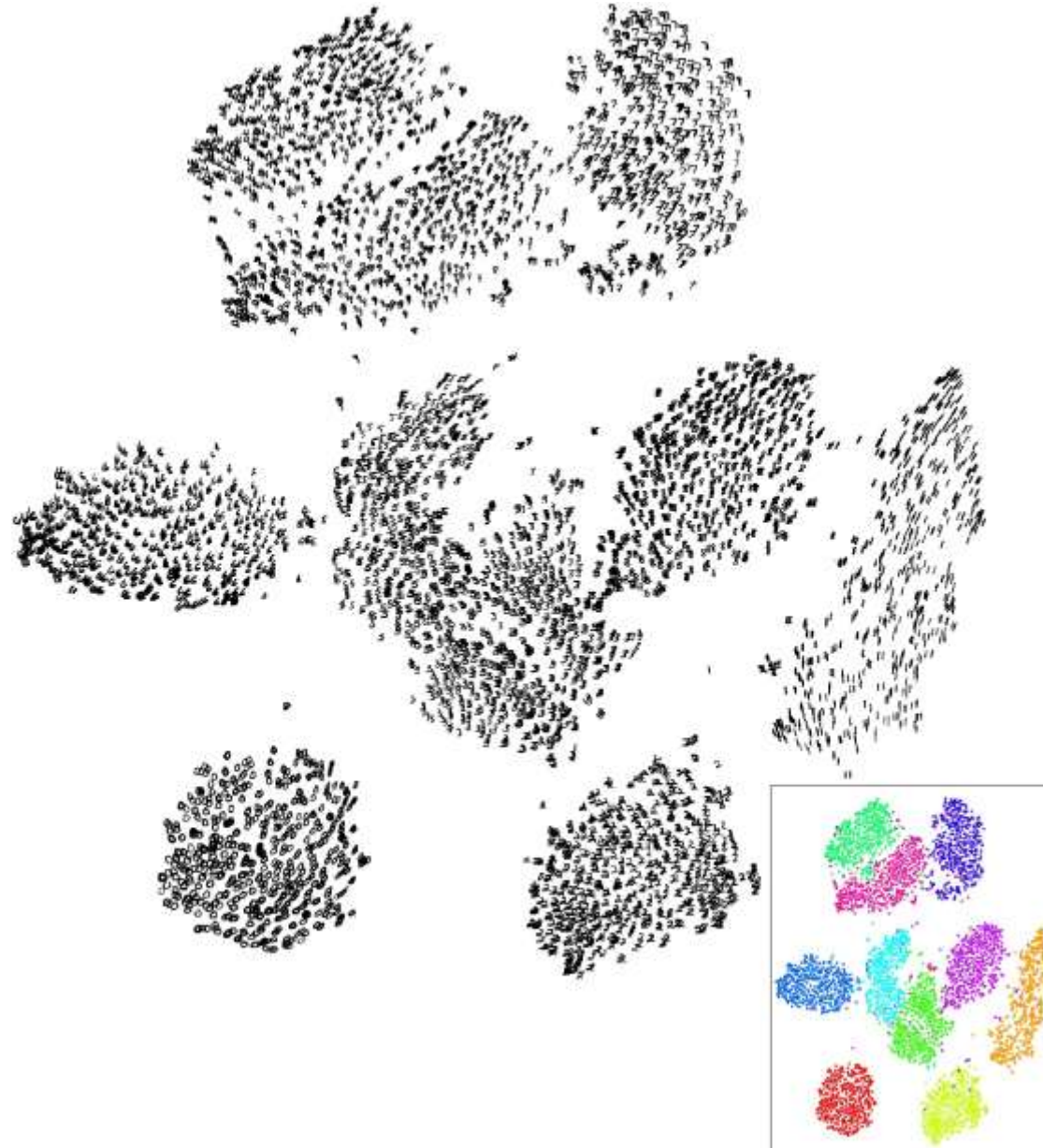
Figure 1:

These images are stored as a 64×400 matrix. Each column of the matrix is an 8×8 greyscale image (the pixel intensities are between 0 and 1).

Representation learning

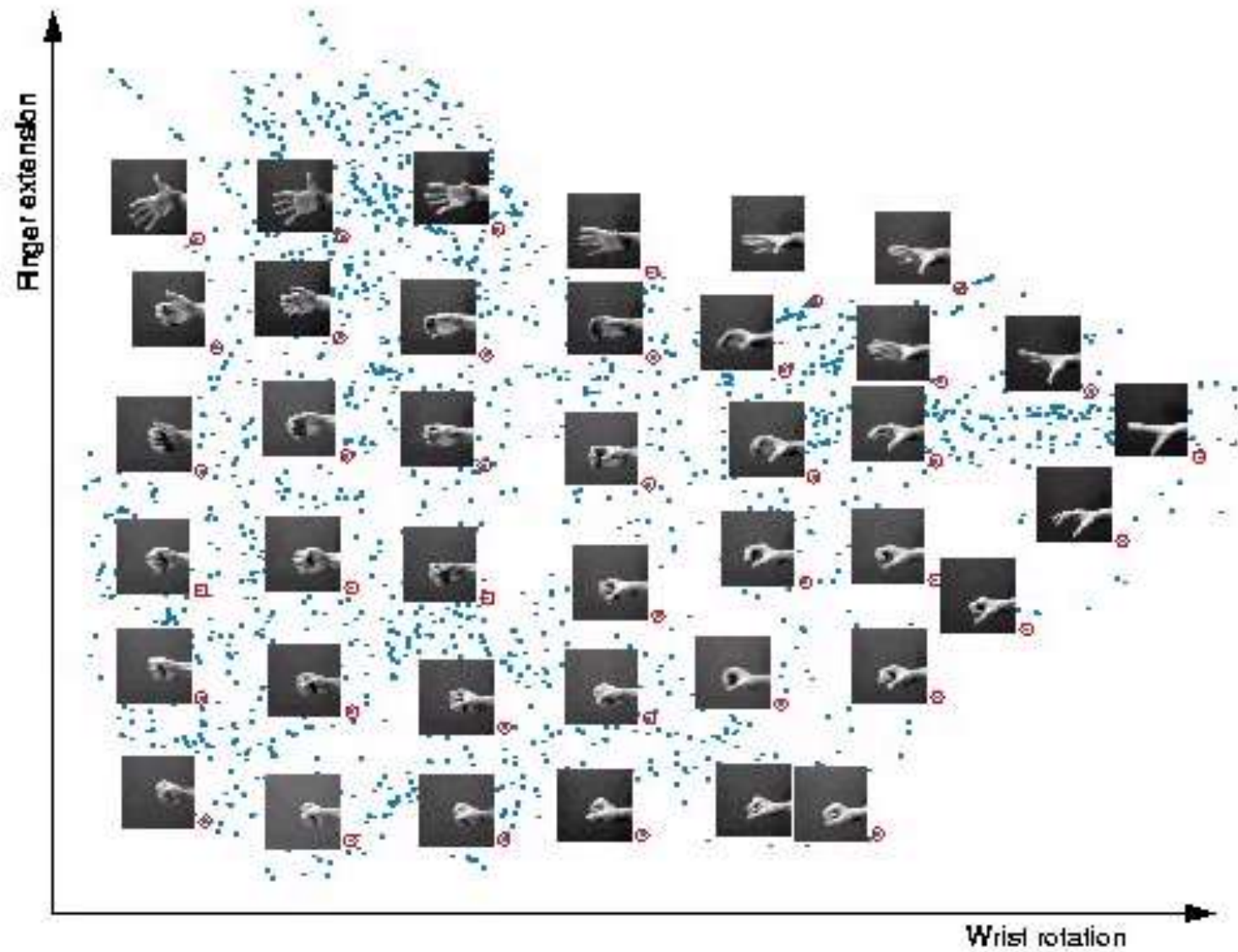


A canonical dimensionality reduction problem from visual perception. The input consists of a sequence of 64-dimensional vectors, representing the brightness values of 8 pixel by 8 pixel images of digits 2 and 3. Applied to $n = 400$ raw images. A two-dimensional projection is shown, with the original input images.



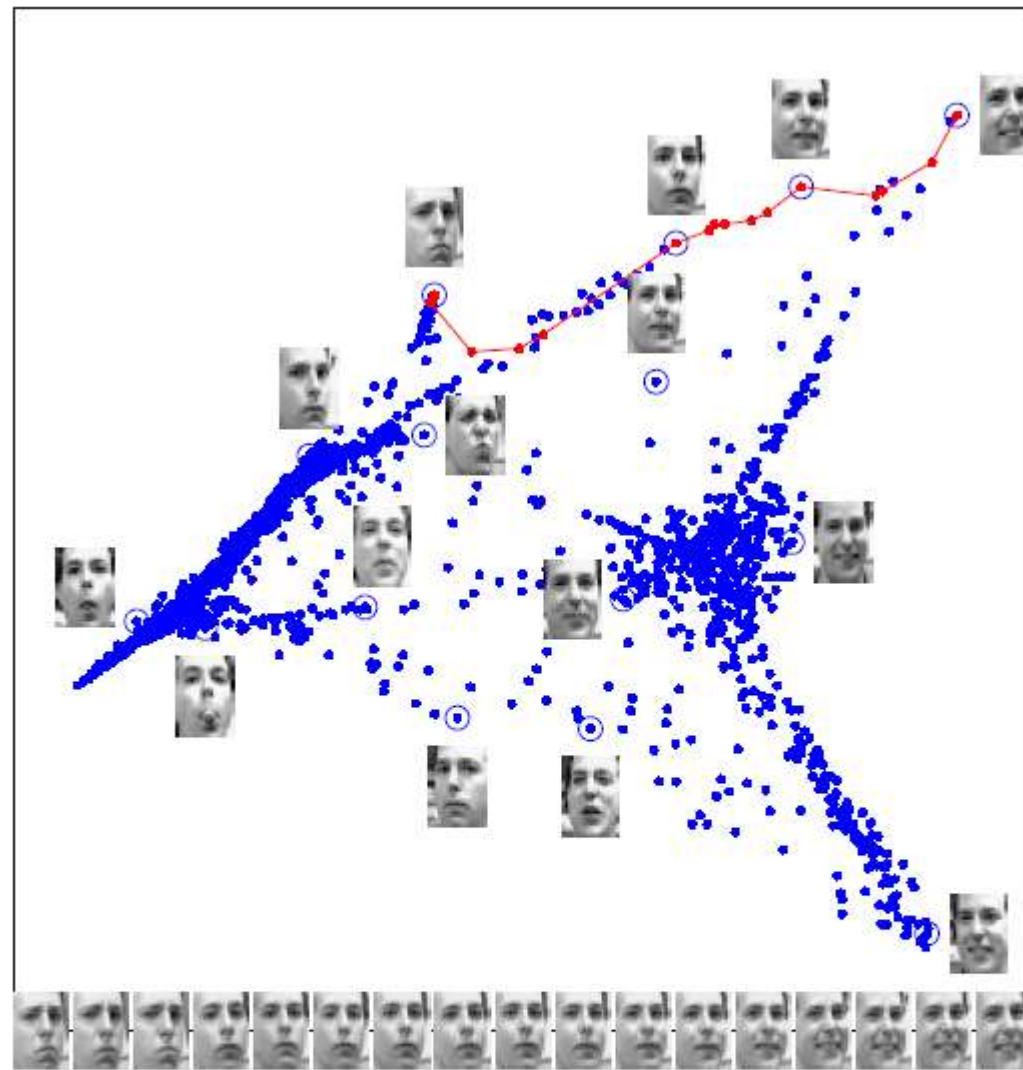


t-SNE: most images of face
the bottom. Most images
clustered on the right.



Example from (Tenenbaum 2000)

Classification, Ali Ghodsi

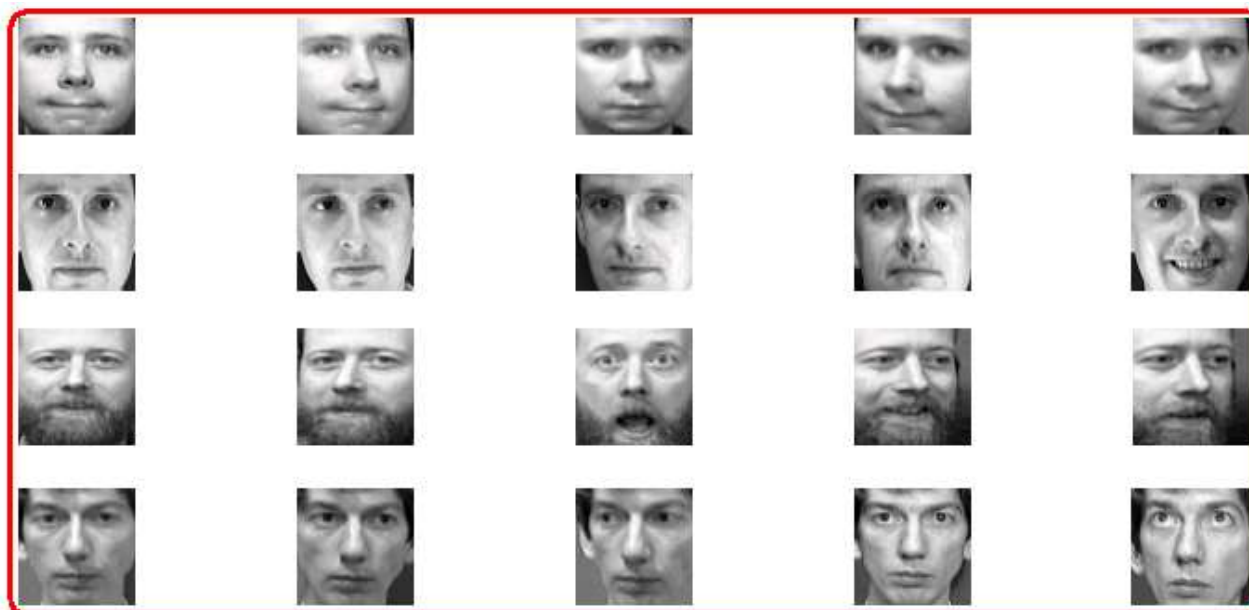


Images of faces mapped into the embedding space described by the first two coordinates of LLE, using $K = 12$ nearest neighbors. Representative faces are shown next to circled points at different points of the space. The bottom images correspond to points along the top-right path, illustrating one particular mode of variability in pose and expression. The data set had a total of $N = 1965$ grayscale images at 20×28 resolution ($D = 560$). Ali Ghodsi

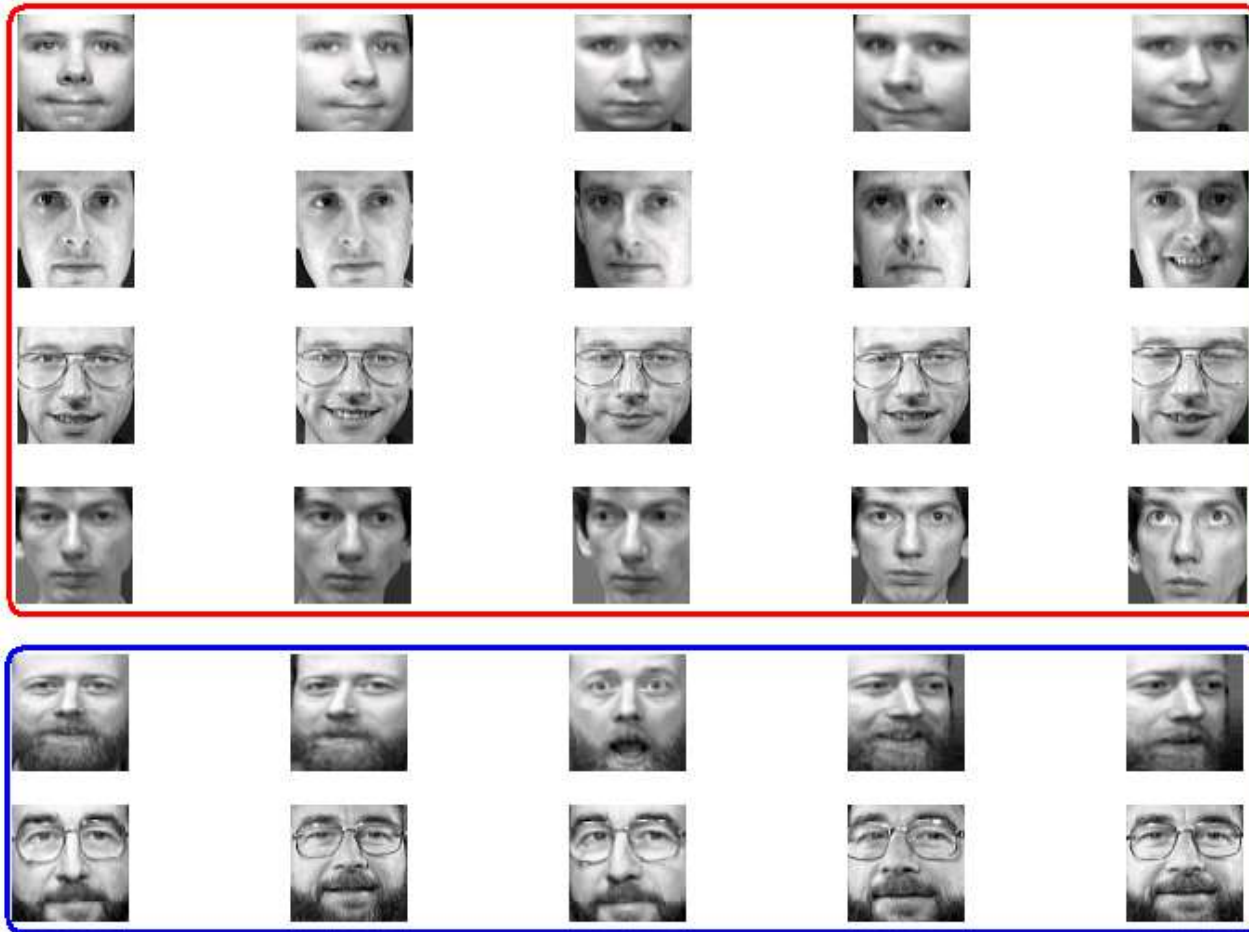
Different Features



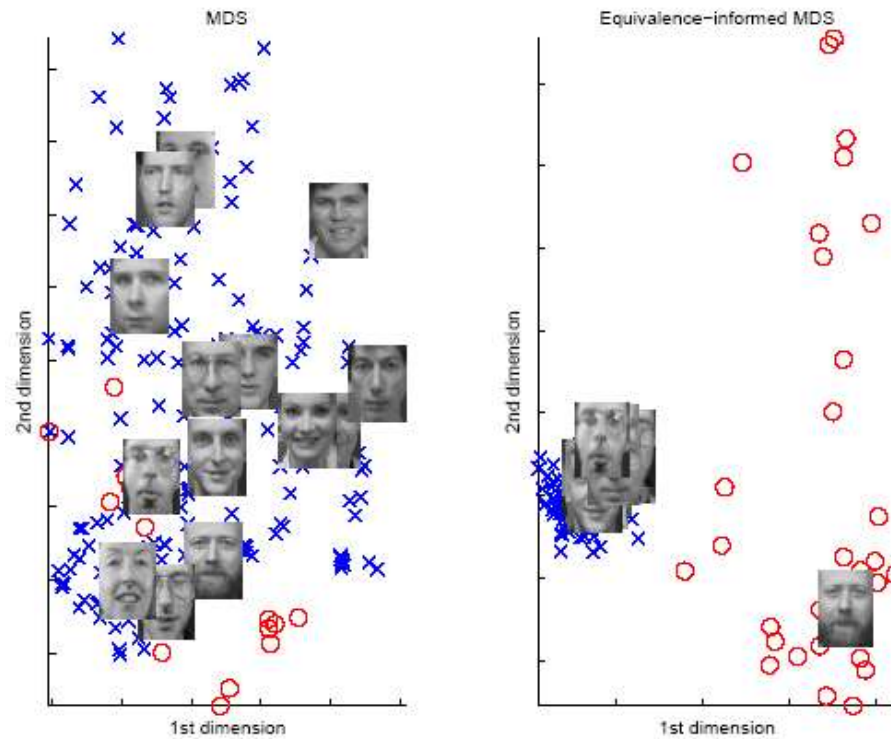
Glasses vs. No Glasses



Beard vs. No Beard

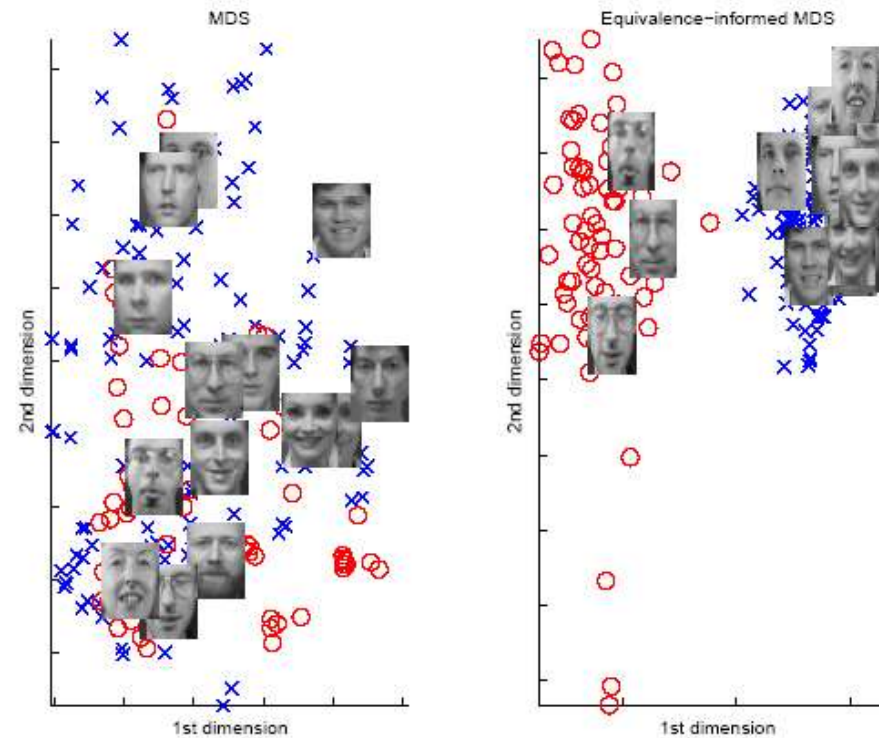


Beard Distinction



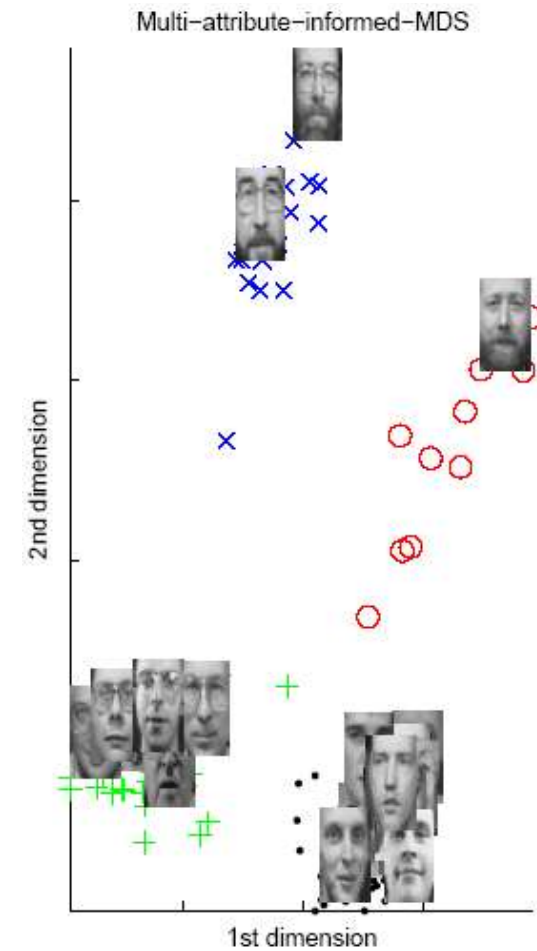
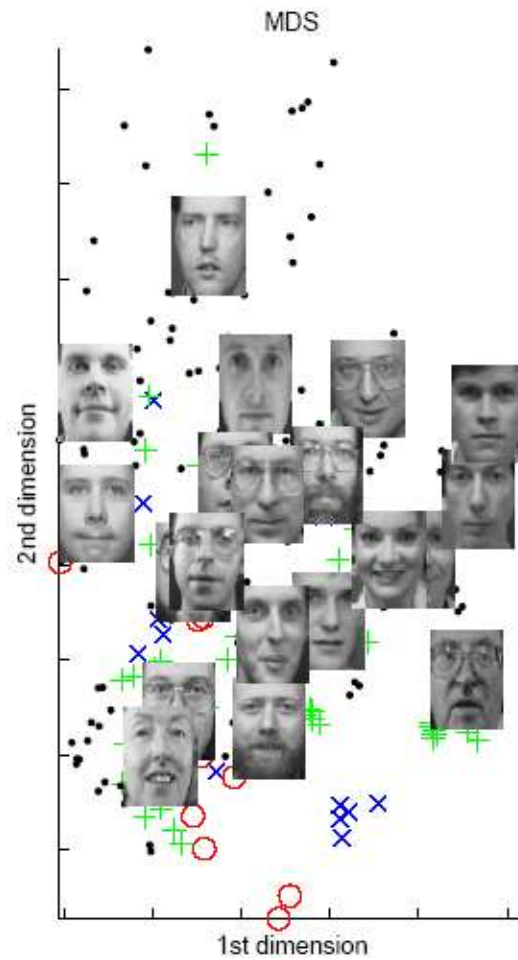
MDS and Equivalence-Informed MDS with bearded/unbearded distinction (all class-equivalent pairs)

Glasses Distinction



MDS and Equivalence-Informed MDS with glasses/no glasses distinction (all class-equivalent pairs)

Multiple-Attribute Metric



Materials and Resources

There is no mandatory textbook. Lectures and posted materials define the course

Recommended references

1. Trevor Hastie, Robert Tibshirani, and Jerome Friedman
The Elements of Statistical Learning
2. Christopher M. Bishop
Pattern Recognition and Machine Learning
3. Kevin P. Murphy
Machine Learning: A Probabilistic Perspective

Assigned readings may also include research papers and instructor-provided notes.

Course Platforms

Platform	Primary role	Access
Course website	Materials, readings, schedules, instructions, and resources	aghodsib.github.io/courses/stat841/
Piazza	Communication, questions, announcements, and private feedback	Widget on the main course webpage
LEARN	Submission dropboxes and grades	Course LMS

Students are responsible for checking all official course platforms regularly.

Communication Policy

Use Piazza for all communication with the instructor and teaching assistant.

PUBLIC POSTS

Use for class-wide questions whose answers may help others.

PRIVATE POSTS

Use for personal matters, private feedback, or questions that reveal assignment or project solutions.

Before posting: search existing threads • use a descriptive subject and relevant tags • keep messages concise and course-related

Assessment and Evaluation

Component	Description	Weight
Assignments	Four individual assignments at 12.5% each	50%
Paper presentation	Small-group presentation of a selected research paper	10%
Group final project	Research or applied classification project	40%
Total		100%

Assignments — 50%

Four individual assignments • 12.5% each

Assignments may include:

- Mathematical derivations and proofs
- Programming and data analysis
- Computational experiments

AUTHORSHIP & UNDERSTANDING

A student may be asked to explain part of a submission orally or complete a short individual quiz. Inadequate demonstrated understanding may reduce or eliminate credit for the affected work.

Submit work that you can explain and defend independently.

Paper Presentation and Final Project

PAPER PRESENTATION — 10%

- Small-group presentation of a selected research paper
- Every member must participate and answer questions
- Group-size limit, venues, selection process, duration, and rubric will be announced

GROUP FINAL PROJECT — 40%

- Substantive classification problem, method, or research question
- Final report required; milestones and criteria will be provided
- Must be new work for this course and cannot reuse work from another course

Tentative Assessment Dates

Assessment	Release	Due
Assignment 1	Mon, Sep 28	Mon, Oct 19 • 11:59 p.m.
Assignment 2	Mon, Oct 19	Mon, Nov 9 • 11:59 p.m.
Assignment 3	Mon, Nov 9	Mon, Nov 23 • 11:59 p.m.
Assignment 4	Mon, Nov 23	Mon, Dec 7 • 11:59 p.m.
Paper presentation	Selection process TBD	Scheduled presentation date
Group final project	—	Mon, Dec 21 • 11:59 p.m.

All dates are tentative and may change. Changes will be announced on official course platforms.

Late Work and Generative AI

LATE & MISSED WORK

Provisional policy for written assignments and the final project:

- Within 24 hours: 10% penalty on the earned mark
- After 24 hours: grade of zero

Confirm final late-work, extension, and accommodation rules before publication.

GENERATIVE AI

- May support learning unless an assessment states otherwise
- Submitted work must reflect the student's own authorship and understanding
- Disclose use and verify all work, citations, code, and results
- Prohibited during oral or quiz-based verification

Unauthorized or undisclosed use may constitute an offence under Policy 71.

Tentative course plan

The order and emphasis may change in response to the pace, needs, and interests of the class.

Tentative topic	Illustrative coverage
Foundations and the Bayes classifier	Supervised learning, loss, risk, decision rules, Bayes error, and error rates
Gaussian and linear classifiers	LDA, QDA, covariance modelling, and discriminant functions
Linear and logistic models	Regression views of classification, likelihood, and optimization
Feature selection and representation	Feature selection, dimensionality reduction, and feature extraction
Evaluation and model selection	Validation, performance measures, complexity, and generalization
Nearest neighbours and metric learning	k-NN, scaling, local decision rules, learned distances, and embeddings
Support vector machines and kernels	Margins, soft margins, duality, kernels, and nonlinear boundaries
Naive Bayes and decision trees	Conditional independence, splitting criteria, pruning, and interpretation
Bagging and boosting	Random forests, variance reduction, adaptive boosting, and additive models
Radial basis function networks	Localized basis functions and classification
Neural networks and deep learning	Feedforward networks, backpropagation, regularization, and deep classifiers
Semi-supervised and emerging methods	Learning with unlabelled data, presentations, and selected research topics

Assignment screening and Turnitin

- Turnitin or other plagiarism-detection software may be used to check the originality and source documentation of submitted work. Submissions to Turnitin are stored on a U.S. server. Students will receive notice of its use and available alternatives; requests for an alternative must be made in the first week or when the relevant assessment details are provided. Waterloo Turnitin information: <https://uwaterloo.ca/academic-integrity/integrity-students/turnitin-andithenticate>